INTERNSHIP TRAINING REPORT

AI & Machine Learning Internship at Pluto Academy

University:	IILM University, Greater Noida	**Student Name:**	Nishant Deshwal
Roll Number:	25scs1003004504	**Program:**	B.Tech CSE (AI & ML)
Session & Sem:	2025-29 | 3rd Semester	**Duration:**	June 2026 (1 Month)
Organization:	Pluto Academy	**Projects:**	Iris Classification & World Happiness EDA

1. Introduction & Objectives

This report details the technical training and practical project execution completed during the one-month Artificial Intelligence and Machine Learning internship at Pluto Academy [cite: 5, 6]. The core objective was to bridge academic curriculum with real-world industry applications by implementing Python data pipelines, Exploratory Data Analysis (EDA), and supervised machine learning classification models [cite: 5, 6].

2. Technology Stack & Tools Used

The projects were executed in Google Colab / Jupyter Notebook environments using industry-standard Python libraries [cite: 5, 6]:

Python **Pandas** **NumPy** **Scikit-Learn** **Matplotlib** **Seaborn** **Google Colab** **Git & GitHub**

3. Project Work & Implementation

Project 1: Iris Flower Classification (Supervised Machine Learning)

Overview: Developed a multi-class classification model to categorize Iris flower species (*Iris setosa*, *Iris versicolor*, and *Iris virginica*) based on sepal and petal measurements (150 total samples) [cite: 5].

- **Data Loading & Inspection:** Loaded dataset from Scikit-Learn into a Pandas DataFrame; confirmed zero missing values and balanced class distribution (50 samples per class) [cite: 5].
- **Exploratory Visualization:** Utilized Seaborn pairplots to analyze feature relationships and class separability [cite: 5].
- **Train-Test Split:** Partitioned the dataset into 120 training samples and 30 testing samples using train_test_split (test_size=0.2) [cite: 5].
- **Model Implementation:** Trained three machine learning algorithms — **Logistic Regression** (max_iter=200), **K-Nearest Neighbors (KNN)** (k=3), and **Random Forest Classifier** (100 estimators) [cite: 5].
- **Evaluation & Results:** All three models achieved 100% accuracy (1.0) [cite: 5]. Evaluated predictions using Confusion Matrix heatmaps, confirming flawless classification performance [cite: 5].

Project 2: World Happiness Report - Exploratory Data Analysis (EDA)

Overview: Conducted comprehensive EDA on the World Happiness Report (2019 dataset, 156 countries) to investigate socio-economic factors influencing national happiness rankings [cite: 6].

- **Data Extraction & Cleaning:** Extracted dataset archive and loaded 2019 CSV using Pandas; verified absolute data integrity with zero missing values [cite: 6].
- **Regional Rankings:** Analyzed the Top 10 happiest countries (led by Finland, Denmark, Norway) and Bottom 10 countries through bar charts [cite: 6].
- **Statistical & Visual Analysis:** Generated scatter plots (GDP vs Happiness Score) and correlation heatmaps across GDP per capita, social support, healthy life expectancy, freedom, generosity, and corruption perceptions [cite: 6].
- **Key Insights:** Discovered that national happiness scores exhibit the strongest positive correlations with GDP per capita (0.79), Healthy Life Expectancy (0.78), and Social Support (0.78) [cite: 6].

4. Key Learnings & Conclusion

The internship at Pluto Academy provided invaluable hands-on experience in end-to-end data science workflows [cite: 5, 6]. By successfully implementing supervised classification algorithms and advanced exploratory data visualizations, technical proficiency in Python, Pandas, Scikit-Learn, and Seaborn was significantly enhanced [cite: 5, 6]. This project work fulfills all requirements for the IILM University internship evaluation [cite: 5, 6].

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